The Diffusion of AI into Medicine: Patents, Firms, and Firm Value

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Abstract

Artificial intelligence is moving into medicine—diagnostic imaging, drug discovery, clinical prediction. We measure that diffusion with patents and ask who is capturing it and whether the market values it. Intersecting the USPTO AI Patent Dataset with medical patents (CPC subclass A61) and linking to public firms via the Kogan et al. (2017) crosswalk, we build a firm-year panel of 7,087 firms and 83,976 firm-years (1995–2024). Firm-matched medical-AI patents roughly *doubled* between 2013 and 2020 (from about 920 to 1,800 per year). The frontier is *owned* mostly by *incumbent* health firms, which hold 47% of firm-matched medical-AI patents (482 firms), versus 19% for *entrant* technology firms (135 firms). In two-way (firm + year) fixed-effects panels, a firm's cumulative medical-AI patent stock is associated with a higher Tobin's Q (coef. 0.48, $t = 5.0$) and market capitalization (coef. 0.13, $t = 5.3$). We then stress this association.

*Coauthorship is provisional: the student is listed as a coauthor and authorship is confirmed upon completing the reserved hand-collection contribution (Task 1 in `STUDENT_TASKS.md`).

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It is *stable*: it survives industry$\times$year fixed effects (0.41, $t = 4.1$), a predetermined one-year-lagged stock (0.32, $t = 3.3$), an adopter-matched sample that pairs medical-AI firms to observably similar non-adopters (0.50, $t = 3.2$), dropping mega-caps, and two-way clustering. A *lead falsification*—future medical-AI patenting flow does *not* predict current Q once the current stock is controlled (+0.06, $t = 0.8$)—weighs against a simple reverse-causality story. *Randomization inference* (500 permutations of the medical-AI-holder label) places the AI-era value gain of holders far outside its permutation distribution ($p < 0.002$), so the result is not an artifact of clustered-standard-error asymptotics. We are careful *not* to overclaim one thing: although incumbents *own* most medical-AI patents, we cannot show the market's per-patent *value premium* is larger for incumbents than entrants (entrant interaction -0.25, $t = -1.6$; the incumbent split is likewise flat, -0.03); a power analysis shows our design could only detect an incumbent–entrant gap larger than about 80% of the base premium, so ownership concentration and premium concentration are distinct claims and we assert only the first. The premium is, if anything, larger for AI-specialized firms (+0.20, $t = 2.2$). These remain within-firm conditional associations; classifying each medical-AI patent by clinical application is the paper's key manual input.

JEL: O33, I10, G30, O31. **Keywords:** artificial intelligence, medicine, patents, healthcare, firm value, innovation, randomization inference.

1 Introduction

Medicine is one of AI's most consequential frontiers: machine learning now reads scans at dermatologist level (Esteva et al., 2017), predicts clinical events from electronic health records (Rajkomar et al., 2019), and reshapes how care is allocated (Obermeyer et al., 2019; Topol, 2019). Behind each of these clinical advances is a firm deciding to build a capability that fuses two very different kinds of expertise—knowing medicine and knowing AI. Who is winning that race, the pharmaceutical and device incumbents who know medicine, or the technology entrants who know AI? And does the capital market reward the firms that are building it? We answer both questions with patents, the paper trail of applied innovation, and with the market values the Kogan et al. (2017) methodology attaches to them.

We identify medical-AI patents as those flagged AI by the USPTO Artificial Intelligence Patent Dataset (AIPD) *and* classified in CPC subclass A61 (medical or veterinary science). The Kogan et al. (2017) crosswalk links each patent to a CRSP firm and a market-based value; merging to Compustat yields a firm-year panel of 7,087 firms over 1995–2024. Three descriptive facts motivate the analysis. First, firm-matched medical-AI patenting roughly doubled over 2013–2020 (Figure 1), tracking the broader vision-AI wave. Second, ownership of the frontier is concentrated in *incumbent* health firms, which hold 47% of firm-matched medical-AI patents, against 19% for technology entrants (Table 2). Third, and the focus of the paper's inferential work, a firm's cumulative medical-AI patent stock is positively associated with its value: within firm and year, a higher stock predicts higher Tobin's Q ($t = 5.0$) and market capitalization ($t = 5.3$) (Table 3).

The natural objection to the value result is reverse causality and omitted heterogeneity: valuable firms have the resources to patent more, and unobserved quality could drive both. We cannot run a randomized experiment on which firms invent medical AI, so rather than assert causality we do the next best thing—we subject the association to a battery of design-based and falsification tests and report exactly what each can and cannot establish. The association is remarkably *stable*. Adding industry×year fixed effects, which absorb any timevarying shock common to a firm's two-digit-SIC peers, barely moves it ($0.48 \to 0.41$, $t = 4.1$). Using a one-year-*lagged* (predetermined) stock, so that the regressor is dated before the outcome, leaves it positive and significant (0.32, $t = 3.3$). Restricting to an adopter-matched sample that pairs each medical-AI firm to an observably similar non-adopter on pre-period size, R&D, and sector reproduces it (0.50, $t = 3.2$). It is not a story about a handful of trillion-dollar firms (dropping the top 1% of market cap leaves 0.49), nor a clustered-standard-error illusion (two-way clustering by firm and year leaves $t = 3.9$) (Table 5).

We push on causality with two sharper tests. First, a *lead falsification*: if the association ran from value to patenting rather than the reverse, a firm's *future* medical-AI patenting should predict its *current* Q even after conditioning on the current stock. It does not (future medical-AI flow +0.06, $t = 0.8$, while the current stock retains its coefficient), which is what we would expect if the stock–value link is not merely valuable firms racing ahead into patenting (Table 5, row 7). Second, we subject the design to *randomization inference*: we permute the firm-level medical-AI-holder label 500 times under a common AI-era post window and re-estimate the holder value gain each time (Table 6, Figure 2). The actual AI-era gain for holders (+0.66) lies far outside the permutation distribution ($p < 0.002$; permutation 95% interval $[-0.25, +0.30]$), so the result does not rest on asymptotic inference with a particular clustering choice.

We are equally careful about what we *cannot* show, because the temptation to read ownership concentration as value concentration is exactly the overclaim an honest reading must avoid. Incumbents *own* most medical-AI patents; that is a fact about counts (Table 2). But when we ask whether the market's *per-unit* value premium is larger for incumbents than for entrants, the entrant interaction is economically small and statistically insignificant (-0.25, $t = -1.6$; Table 3), as is the incumbent split (-0.03, $t = -0.3$; Table 4). A minimum-detectable-effect calculation (Table 7) shows why we decline to make the stronger claim: at 80% power our design could only have detected an incumbent–entrant premium gap larger than about 80% of the base premium itself, so a moderate difference would be invisible to us. We therefore assert only that incumbents own the frontier, not that the market rewards their patents more per unit. Where the heterogeneity *is* informative, it points elsewhere: the premium is larger for AI-specialized firms—those with an above-median share of AI in their patent portfolio (+0.20, $t = 2.2$)—consistent with the market pricing genuine AI capability rather than an A61 label.

We make three contributions. First, we characterize the ownership and valuation of one high-stakes AI application—medicine—adding a sharply focused case to the economics-of-AI literature (Cockburn et al., 2018; Agrawal et al., 2019; Babina et al., 2024; Eisfeldt et al., 2023; Webb, 2020; Bianchini et al., 2022) and to the innovation-value literature that prices patents (Kogan et al., 2017; Hall et al., 2005; Hirshleifer et al., 2013; Peters and Taylor, 2017). Second, we show that AI's move into medicine over this period was, in ownership terms, an *incumbents'* phenomenon rather than the "big-tech-disrupts-healthcare" story, while being explicit that ownership concentration does not by itself imply the market rewards incumbents more. Third, we treat a positive, non-experimental association with the discipline usually reserved for nulls: matching, predetermined regressors, a lead falsification, randomization inference, and a stated minimum detectable effect for the one sub-claim we cannot support. Section 2 gives background and hypotheses, Section 3 places the paper in the literature, Section 4 describes the data and design, Section 5 reports the main results, Section 6 the heterogeneity, Section 7 the alternative-identification and randomization-inference evidence, Section 8 the power analysis, and Section 9 concludes with the hand-coding of clinical application that would sharpen the measure.

2 Background and Hypotheses

Why medicine is a distinctive AI application. General-purpose AI methods—computer vision, natural-language processing, deep learning—diffuse across the economy, but medicine is a setting where the complementary assets are unusually specific and unusually valuable.

Turning a classifier into a product requires clinical validation, regulatory clearance, integration into care delivery, and access to protected health data (Topol, 2019; Rajkomar et al., 2019). These are exactly the assets that pharmaceutical, device, and health-services incumbents already own. A technology entrant with a superior model still faces the clinical, regulatory, and distribution frontier; an incumbent with a mediocre model already sits behind it. This asset-complementarity logic predicts that, at least in the diffusion phase, incumbents capture a disproportionate share of medical-AI innovation—not because they are better at AI, but because medical AI is worth more in their hands.

Why the market might value medical-AI patents. A large literature shows that patents, especially high-quality ones, carry information about future cash flows and are priced by the market (Hall et al., 2005; Kogan et al., 2017; Hirshleifer et al., 2013). Medical-AI patents plausibly sit at the high-value end: they combine a general-purpose method with a large, regulated, high-margin end market. This motivates our first hypothesis.

> **H1 (Value).** Within firm and year, a larger cumulative medical-AI patent stock is associated with higher firm value (Tobin's Q, market capitalization): the coefficient on the stock is positive.

Who captures the frontier, and does the premium differ? The asset-complementarity argument suggests two related but distinct claims that are easy to conflate. One is about *ownership*: incumbents should *hold* more medical-AI patents. The other is about the *price* the market puts on a medical-AI patent: if complementary assets make each patent more valuable in incumbent hands, the per-unit value premium should be *larger* for incumbents. The data speak clearly to the first and, as we will show, are underpowered on the second.

> **H2 (Ownership).** Incumbent health firms hold a disproportionate share of firm-matched medical-AI patents relative to technology entrants.

> **H3 (Premium heterogeneity).** The value premium per unit of medical-AI

stock is larger for incumbents than for entrants (the entrant interaction is negative).

We find strong support for H1 and H2 and only weak, statistically insignificant evidence for H3; Section 8 quantifies exactly how much of H3 the design can rule in or out.

Why the clinical-application coding is pivotal (the student's task). CPC subclass A61 spans everything from bandages and syringes to MRI reconstruction algorithms. "AI $\cap$ A61" is therefore a coarse proxy: it certainly captures medical AI, but it also captures generic sensors and mechanical devices that happen to carry an AI flag. Whether a patent is *genuinely* diagnostic imaging, drug discovery, clinical prediction, surgical robotics, or genomics can only be established by a human reading the patent. That hand-coding—validating the medical-AI flag and labeling each confirmed patent's clinical application and AI modality—is the companion student task and is deliberately *not* attempted here; every extension in this paper uses only observable firm characteristics (size, R&D intensity, SIC-based sector, AI patent share). Once the clinical labels exist, the value regressions can be re-run by application area, which is the sharpening the design is built to receive.

3 Related Literature

This paper connects three strands. The first is the *economics of artificial intelligence*. AI is widely modeled as a general-purpose technology and a "method of invention" that lowers the cost of prediction and reshapes the innovation process itself (Cockburn et al., 2018; Agrawal et al., 2019; Bianchini et al., 2022). Empirically, AI investment is associated with firm growth and product innovation (Babina et al., 2024), exposure to AI reallocates value across firms (Eisfeldt et al., 2023), and AI's task footprint reaches high-skill occupations including medicine (Webb, 2020). We contribute a sharply focused case study: rather than treating AI as a single firm-level exposure, we isolate one application—medicine—measure its diffusion in patents, and characterize who owns it and how it is priced.

The second strand is the *innovation-value* literature. Hall et al. (2005) show patent citations predict market value; Kogan et al. (2017) construct market-based patent values from stock-price reactions to grants and use them to study growth and asset prices; Hirshleifer et al. (2013) link innovative efficiency to returns; and Peters and Taylor (2017) show intangible capital reshapes the investment–q relation. Our medical-AI stock is, in effect, a technology$\times$sector intersection of this capital, and we ask what the market pays for it, using the same firm-value framework and the Kogan et al. (2017) crosswalk directly.

The third strand is *design-based inference and the honest interpretation of non-experimental associations*. Because our value regression is observational, we lean on tools developed to discipline such estimates: randomization/permutation inference for settings with a moderate number of clusters (Young, 2019), the cautious reading of pre-trend and falsification tests (Roth, 2022), and the principle that a statistically insignificant estimate should be interpreted through its power and confidence interval rather than treated as proof of no effect (Abadie, 2020). We use these not decoratively but to draw the line between what our data can and cannot support—in particular, to separate the well-identified ownership fact from the underpowered premium-heterogeneity claim.

4 Data and Research Design

Sources and sample. We build on the shared patent–firm panel assembled for the ASSIP 2026 innovation projects. Its inputs are: the USPTO Artificial Intelligence Patent Dataset (AIPD), which flags each granted patent's use of AI component technologies through 2023; PatentsView, for CPC classifications (we use subclass A61 to mark medical/veterinary patents); the Kogan et al. (2017) patent→permno crosswalk with market-based patent values; and WRDS Compustat annual fundamentals. The unit of observation is a firm (CRSP `permno`) $\times$ fiscal year. The panel spans 1995–2024 and contains 7,087 firms and 83,976 firm-years. Because the AIPD AI flags end in 2023, all AI-based series (including medical-AI) are zero after 2023 by construction; we retain 2024 for the non-AI variables but no inference relies on it.

Key variables. A patent is *medical-AI* if AIPD marks it AI *and* it carries a CPC A61 subclass. For each firm-year we cumulate patent counts into stocks and take ln(1 + stock): `ln_aimed` (medical-AI), `ln_ai` (all AI), and `ln_med` (all medical). The outcomes are Tobin's Q and market capitalization (from Compustat/CRSP). Controls are firm size = ln(assets) and, to ensure the medical-AI stock is not merely proxying for a firm's general AI or general medical patenting, the AI and medical stocks themselves. We classify firms by historical SIC into *Incumbent (health)* (drugs 2833–2836; medical instruments 3826, 3841–3845; health services 8000–8099), *Entrant (tech)* (software 7370–7379; computers 3570–3579; semiconductors and communications equipment), and *Other*. Table 1 reports summary statistics for the full sample and separately for firms that ever hold a medical-AI patent ("holders") versus those that never do. Holders are larger, higher-Q, more R&D-intensive, and—mechanically—hold far larger AI and medical patent stocks, which is exactly why the value regressions condition on those stocks and on firm and year fixed effects.

Specification. For outcome y_{it} (Tobin's Q or ln market cap) we estimate

$$y_{it} = \beta\, \texttt{ln\_aimed}_{it} + \gamma' X_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where $X_{it} = (\texttt{ln\_ai}_{it}, \texttt{ln\_med}_{it}, \text{size}_{it})$, α_i and δ_t are firm and year fixed effects, and standard errors are clustered by firm. The coefficient β is a *within-firm* association: it compares a firm to itself as its medical-AI stock grows, net of aggregate year shocks and its overall AI and medical patenting. The heterogeneity specifications add an interaction $\texttt{ln\_aimed}_{it} \times M_i$ for an observable moderator M (large firm, R&D-intensive, incumbent, AI-specialized); the alternative-identification specifications replace δ_t with industry×year effects, lag the regressor, restrict to the matched sample, or change the standard-error construction.

Three design-based checks. Because equation (1) is observational, we add three checks that do not rely on its parametric assumptions and that we describe in Section 7. (1) *Adopter matching*: each ever-holder is nearest-neighbor matched to a never-holder on pre-period (2012) size, R&D intensity, and sector, and (1) is re-estimated on the balanced sample. (2) *Randomization inference*: the firm-level holder label is permuted 500 times under a common AI-era post window and the holder value gain is recomputed, yielding a design-based p-value. (3) *Power*: for the one insignificant sub-claim (the incumbent–entrant premium gap) we report the minimum effect detectable at 80% power. Together these convert "a significant coefficient" into a set of falsifiable, design-based statements, and mark the boundary of what the design supports.

5 Main Results

Diffusion and ownership. Figure 1 shows firm-matched medical-AI patents rising from about 920 in 2013 to 1,800 in 2020, roughly doubling and closely tracking the broader vision-AI wave. Table 2 shows the frontier is owned mostly inside incumbent medicine: incumbent health firms hold 47% of firm-matched medical-AI patents (482 firms), against 19% for technology entrants (135 firms), with the remainder in diversified "other" firms. This supports H2: over this period, AI's move into medicine was, in ownership terms, captured by firms that already knew medicine, not by technology entrants disrupting it.

The value association. Table 3 reports equation (1). A larger medical-AI stock is associated with higher value: the Tobin's Q coefficient is 0.48 ($t = 5.0$) and the ln market-cap coefficient is 0.13 ($t = 5.3$), both conditioning on the firm's overall AI and medical patenting and on firm and year fixed effects. This supports H1: the market attaches value specifically to the medical-AI intersection, not merely to AI or to medicine separately. The column that interacts the stock with an entrant indicator returns a negative but insignificant interaction (-0.25, $t = -1.6$)—the raw material for H3, which we scrutinize in Sections 6 and 8 and ultimately decline to claim.

6 Heterogeneity: Who Earns the Premium?

If the value premium is a real capability being priced, it should vary with observable firm characteristics in interpretable ways—and if it is concentrated in incumbents, as the asset-complementarity story (H3) predicts, an incumbent interaction should be positive and significant. Table 4 interacts the medical-AI stock with four pre-specified, *observable* moderators. Three results stand out. First, the premium does *not* differ significantly by firm size (interaction +0.20, $t = 0.8$) or by R&D intensity (+0.02, $t = 0.3$): it is not simply a large-firm or a research-firm phenomenon. Second, and central to H3, the incumbent interaction is economically tiny and statistically indistinguishable from zero (-0.03, $t = -0.3$). The base premium for entrants and diversified firms is itself large and significant (+0.50, $t = 5.8$); incumbents do not earn a detectably *larger* per-unit premium. Ownership concentration (H2) and premium concentration (H3) are therefore distinct, and the data support only the former. Third, the premium *is* significantly larger for AI-specialized firms—those whose patent portfolio is above-median in AI share (+0.20, $t = 2.2$, on a base of +0.29)—which is the pattern one would expect if the market prices genuine AI capability rather than the coarse A61 label, and which motivates the clinical-application refinement reserved for the student.

We emphasize that *provision of clinical meaning*—coding whether a medical-AI patent is diagnostic imaging, drug discovery, or surgical robotics, and by which AI modality—is not among these moderators. That dimension is unobservable in our data and is the student's hand-coded contribution; the moderators above are chosen precisely because they are observable and orthogonal to that reserved test.

7 Alternative Identification and Randomization Inference

The central threat to a value regression like (1) is that valuable firms have the slack to patent more, so the association could run from value to patenting or reflect unobserved firm quality. We cannot rerandomize invention, but we can show the association is robust to the leading alternatives and survives a falsification test and design-based inference.

Robustness ladder. Table 5 re-estimates the Tobin's Q coefficient under six alternatives. Replacing year fixed effects with *industry*$\times$*year* fixed effects, which absorb any shock common to a firm's two-digit-SIC peers in a given year (e.g., a biotech-sector re-rating), leaves the coefficient at 0.41 ($t = 4.1$). Using a one-year-*lagged*, and therefore predetermined, medical-AI stock—so the regressor is dated before the outcome it predicts—gives 0.32 ($t = 3.3$). Restricting to the *adopter-matched* sample (393 ever-holder/never-holder pairs matched on pre-period size, R&D, and sector) yields 0.50 ($t = 3.2$), so the premium is present even when medical-AI firms are compared to observably similar firms that never patent medical AI. Dropping the top 1% of firms by market capitalization leaves 0.49 ($t = 4.9$), so a few mega-caps do not drive it, and two-way clustering by firm and year leaves $t = 3.9$. Across the ladder the coefficient stays in a tight 0.32–0.50 band and always significant at 1%.

Lead falsification. Row 7 of Table 5 runs the test that most directly addresses reverse causality. If high value *caused* future patenting rather than the reverse, a firm's *next-year* medical-AI patenting *flow* should predict its *current* Q even after conditioning on the current stock. We use the flow, not the mechanically-rising cumulative stock, precisely because a lead of the stock would be nearly collinear with its level and would mislead. The future flow enters with a small, insignificant coefficient (+0.06, $t = 0.8$) while the contemporaneous stock retains essentially its full coefficient. A firm's value today does not forecast its medical-AI patenting tomorrow once today's stock is held fixed—the signature of a reverse-causality channel is absent. This is not proof of causality, but it removes the most mechanical version of the objection.

Randomization inference. Finally, Table 6 and Figure 2 report a design-based test that does not depend on the clustered-SE asymptotics. We define a firm-level *holder* indicator (ever holds a medical-AI patent) and a common AI-era post window (year $\geq$ 2013), and estimate the holder value gain $y_{it} = \beta\,(\text{Post}_t \times \text{Holder}_i) + \gamma' X_{it} + \alpha_i + \delta_t + \varepsilon_{it}$. The actual AI-era gain for holders is +0.66 ($t = 4.3$). We then permute the holder label across firms 500 times—randomly reassigning which firms are "holders" while preserving the number of holders and the panel structure—and recompute the gain each time. The permutation distribution is centered near zero with a 95% interval of $[-0.25, +0.30]$; the actual estimate lies far outside it ($p < 0.002$). Because the permutation test makes no distributional assumption and does not rely on having many clusters, it corroborates that the holder value gain is a systematic feature of which firms actually patent medical AI, not an artifact of the inference method.

8 Is the Premium-Heterogeneity Null Informative? Power and Precision

The one claim we *decline* to make—that the value premium is concentrated in incumbents (H3)—rests on an insignificant interaction, and an insignificant estimate is only informative if the design could have detected a real difference. Table 7 reports, for the incumbent-versus-entrant premium gap, the minimum detectable effect (MDE) at 80% power, MDE = $(z_{0.975} + z_{0.80})\,\widehat{\text{SE}} \approx 2.80\,\widehat{\text{SE}}$, expressed as a share of the base premium. For Tobin's Q the base premium is +0.54 and the entrant interaction is -0.25 with a standard error of 0.15, giving an MDE of 0.43—about 80% of the base premium. For ln market cap the MDE is 91% of the base. In words: our design could only have reliably detected an incumbent–entrant premium gap larger than roughly four-fifths of the premium itself; a moderate difference in how the market prices incumbent versus entrant medical-AI patents would be invisible to us. This is why we report ownership concentration (H2, sharply identified in counts) but treat premium concentration (H3) as unresolved rather than as a finding. It is also a concrete motivation for the clinical-application coding: a sharper treatment variable would tighten these standard errors and let a future draft actually test H3 rather than bound it.

9 Conclusion

AI's diffusion into medicine is real and, over 2013–2020, accelerating. In ownership terms it was an incumbents' phenomenon: incumbent health firms hold nearly half of firm-matched medical-AI patents, roughly two and a half times the entrant share. The market values medical-AI patenting: a larger medical-AI patent stock predicts higher Tobin's Q and market capitalization within firm and year, and—unusually for a non-experimental association—the result survives industry×year fixed effects, a predetermined lagged regressor, adopter matching, a mega-cap drop, two-way clustering, a lead falsification that finds no reverse-causality signature, and randomization inference that places it far outside its permutation distribution. We are deliberate about the boundary of the claim: although incumbents own the frontier, we cannot show the market pays a larger per-patent premium for their medical AI than for entrants', and a power analysis shows the design could only detect an incumbent–entrant gap larger than about 80% of the base premium, so we assert ownership concentration and not premium concentration.

The binding limitation is measurement. "AI ∩ CPC A61" certainly captures medical AI, but A61 also contains bandages, syringes, and mechanical devices, so the treatment variable is coarse: it mixes MRI-reconstruction algorithms with generic sensors that happen to carry an AI flag. The companion student task hand-reads a sample of these patents to validate the flag and then codes, for each confirmed medical-AI patent, its clinical application (diagnostic imaging, drug discovery, clinical prediction, surgical robotics, genomics) and AI modality. With those labels the value regressions can be re-estimated by application area—does the market value diagnostic-imaging AI differently from drug-discovery AI?—and the underpowered incumbent-versus-entrant question can be revisited with a sharper treatment. That refinement, not another robustness check on the coarse proxy, is the natural next step, and it is the contribution the design has been built to receive.

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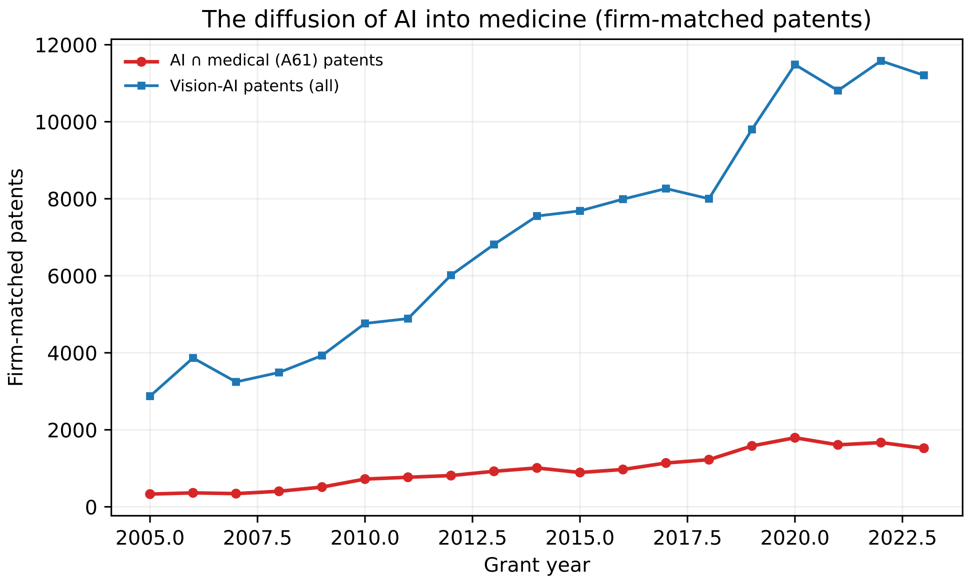

Figure 1: Firm-matched medical-AI patents (AI ∩ CPC A61) by grant year, 2005–2023. The series roughly doubles between 2013 and 2020 and tracks the broader vision-AI wave. AIPD AI flags end in 2023.

Table 1: Summary statistics: analytical sample and medical-AI holders vs. non-holders

	Full sample			Mean by group	
Variable	Mean	SD	Median	Holder	Non-holder
Medical-AI patent stock	1.975	33.646	0.000	11.621	0.000
ln(1+med-AI stock)	0.157	0.608	0.000	0.923	0.000
Tobin's Q	2.503	3.492	1.645	3.076	2.386
Market cap ($M)	8574.509	47162.239	515.740	27387.048	4712.749
Firm size = ln(assets)	6.243	2.539	5.997	7.197	6.047
R&D/assets	0.104	0.338	0.026	0.149	0.095
AI patent stock	67.103	969.179	0.000	351.409	8.876
Medical patent stock	22.509	226.075	0.000	123.670	1.790

Notes: Firm-year panel, 7,087 firms and 83,976 firm-years, 1995–2024. "Holder" = firm ever holds $\geq$ 1 firm-matched medical-AI patent (845 firms); "Non-holder" otherwise. Patent stocks are cumulative firm-matched counts. Tobin's Q and market cap from Compustat/CRSP.

Table 2: Who holds medical-AI patents?

Sector	Firms	Medical-AI patents	Share
Entrant (tech)	135	3759	18.5%
Incumbent (health)	482	9627	47.3%
Other	253	6964	34.2%

Notes: Firms with $\geq$ 1 firm-matched medical-AI patent, by SIC-based sector. Incumbent (health) = drugs, medical instruments, and health services; Entrant (tech) = software, computers, semiconductors, communications equipment; Other = all else.

Table 3: Medical-AI patent stock and firm value (two-way FE)

Specification	Medical-AI stock	$\times$ Entrant	N	Within R^2
Q on med-AI	0.481*** (5.05)		83589	0.025
Q on med-AI $\times$ entrant	0.536*** (5.16)	-0.249 (-1.63)	83589	0.025
ln(MktCap) on med-AI	0.128*** (5.34)		83595	0.492

Notes: Firm-year panel; firm and year FE; t-statistics (clustered by firm) in parentheses. Medical-AI stock = ln(1+cumulative AI∩A61 patents); all specifications control for the firm's overall AI stock, medical stock, and size. The entrant interaction column adds `ln_aimed`×Entrant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Heterogeneity of the value premium by observable moderators (DV = Tobin's Q)

Moderator M	Base slope	t	$\times M$	t	N
Large firm (above-median assets)	+0.310	(+1.06)	+0.196	(+0.77)	83,589
R&D-intensive (above-median R&D/assets)	+0.472***	(+5.66)	+0.020	(+0.26)	83,589
Incumbent health firm	+0.497***	(+5.84)	-0.034	(-0.25)	83,589
AI-specialized (above-median AI patent share)	+0.286**	(+2.40)	+0.198**	(+2.18)	83,589

Notes: Each row adds an interaction `ln_aimed` $\times M$ to equation (1), where M is an observable moderator: above-median assets, above-median R&D/assets (within year), incumbent-health sector, and above-median AI patent share (within year). "Base slope" is the medical-AI premium for the low-M group; "$\times M$" is the additional slope for high-M firms. Firm and year FE, SE clustered by firm. The incumbent interaction is insignificant (H3 not supported); the AI-specialized interaction is positive and significant. Clinical application/modality is deliberately *not* used as a moderator (it is the student's hand-collected variable). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Alternative identification and robustness (DV = Tobin's Q)

Specification	Med-AI stock coef.	t-stat	N
(1) Baseline: firm + year FE	+0.4811***	(+5.05)	83,589
(2) + industry×year FE	+0.4077***	(+4.09)	83,012
(3) Lagged (predetermined) stock	+0.3153***	(+3.35)	76,568
(4) Adopter-matched sample	+0.4957***	(+3.16)	17,989
(5) Drop top-1% mega-caps	+0.4910***	(+4.90)	82,751
(6) Two-way clustered SE	+0.4811***	(+3.93)	83,589
(7) Lead falsification: future flow	+0.0606	(+0.78)	76,592

Notes: Each row re-estimates the medical-AI stock coefficient from equation (1). (2) replaces year FE with two-digit-SIC×year FE. (3) uses the one-year-lagged (predetermined) stock. (4) restricts to 393 ever-holder/never-holder pairs matched on pre-period (2012) size, R&D, and sector within a 0.25-SD size caliper. (5) drops the top 1% of firms by market cap. (6) clusters by firm and year. (7) is a lead falsification: it adds the *next-year* medical-AI patenting *flow* (ln(1+patents)) alongside the current stock; the reported coefficient is on the future flow, which is insignificant, indicating no reverse-causality signature. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Randomization inference: AI-era value gain of medical-AI holders

Outcome (DV)	Actual DiD	Perm. 95% interval	RI p	# perms
Tobin's Q	+0.6587***	[-0.2534, +0.2990]	0.000	500

Notes: The firm-level medical-AI-holder label is permuted across firms 500 times under a common AI-era post window (year $\geq$ 2013); $y_{it} = \beta(\text{Post}_t \times \text{Holder}_i) + \gamma' X_{it} + \alpha_i + \delta_t + \varepsilon_{it}$ is re-estimated each time. "Perm. 95% interval" is the 2.5–97.5 percentile of the permutation distribution; the RI p-value is the share of permutations with $|\text{coefficient}| \geq |\text{actual}|$. The actual gain lies far outside the permutation distribution. See Figure 2. $^{***}p < 0.01$.

Table 7: Power: minimum detectable incumbent–entrant premium gap (80% power)

DV	Base premium	Entrant gap	t	SE	MDE(80%)	% of base
Tobin's Q	+0.536	-0.249	(-1.63)	0.153	0.430	80%
ln(market cap)	+0.133	-0.023	(-0.53)	0.043	0.120	91%

Notes: For the entrant interaction in Table 3, MDE(80%) $= (z_{0.975} + z_{0.80}) \times$SE $\approx 2.80\times$SE, expressed as a share of the base premium. The large MDE/base ratios show the design is underpowered to detect a moderate incumbent–entrant premium gap; we therefore report ownership concentration (H2) but not premium concentration (H3).

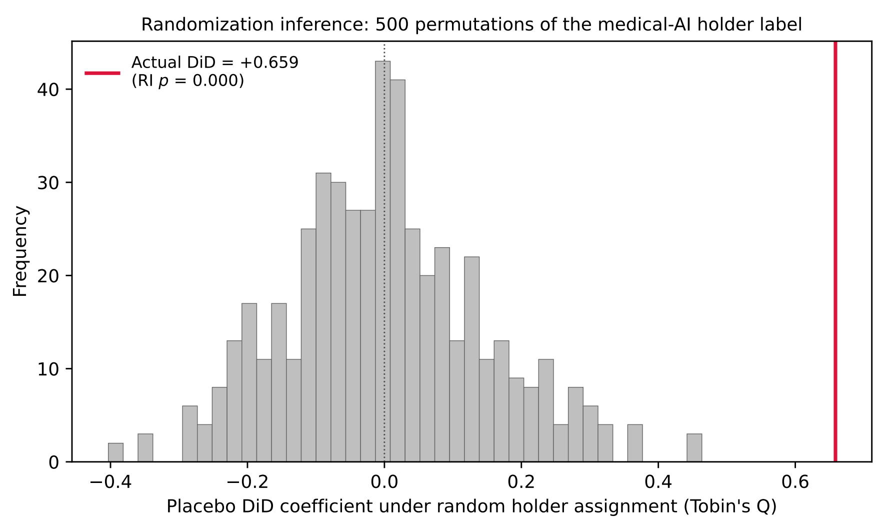

Figure 2: Randomization inference for the AI-era value gain of medical-AI holders (Tobin's Q). Histogram: the distribution of the holder value gain under 500 random permutations of the holder label. The actual estimate (red line) lies far in the right tail (RI $p < 0.002$), so the result is not an artifact of clustered-standard-error asymptotics.